Appendix

1 Appendix

1.1 Runtime Profiling and Policy Implementation

Offline resource profiling. For each supported roadmap level, RAS-FL physically constructs the corresponding compact model and records its active FLOPs, compact-state size, resident model memory, and peak memory. The structure-removal percentage is used only to generate candidate operating points; runtime selection is based on their measured resource profiles. Parameter count is reported as an architecture-level efficiency metric but is not required as an online constraint. FLOPs provide a hardware-independent proxy for computational demand, whereas memory usage is profiled separately for each evaluated hardware and software environment.

Feasible configurations. Given the available resource vector $\mathbf{r}_u(t)$, the feasible configuration set is

$$\mathcal{F}_u(t) = \left\{ p \in \{0\} \cup \mathcal{P} : \mathbf{c}^p_{u,j} \leq \mathbf{r}_{u,j}(t), \quad \forall j \in \mathcal{J}_u(t) \right\}. \tag{1}$$

By default, $\widehat{A}^p$ denotes the server-side validation accuracy associated with configuration p and stored with the global roadmap. The runtime policy therefore selects

$$p_u(t) = \arg \max_{p \in \mathcal{F}_u(t)} \widehat{A}^p. \tag{2}$$

Ties are resolved in favour of the configuration with lower active FLOPs. If $\mathcal{F}_u(t) = \emptyset$, the client invokes the deployment system's fallback procedure rather than executing an infeasible configuration.

Configuration switching. A switch is performed only when the newly selected configuration differs from the currently active one. The client releases the active accelerator model, constructs the selected compact model on the CPU, and transfers it to the accelerator. The new configuration remains active until a change in the resource state causes the policy to select another operating point.

Runtime resource evaluation. We evaluate the supported roadmap configurations under full, moderate, and constrained resource budgets. For each configuration, we report the selected shrinkage level, predictive accuracy, active FLOPs, compact-state size, resident model memory, and peak memory. These measurements demonstrate that the runtime policy can select a feasible operating point as the available resource budget changes.

1.2 Optional Runtime Roadmap Personalization

RAS-FL uses the global roadmap $\mathcal{R}^g$ by default. When a deployment client u has a local adaptation set $\mathcal{D}_u^a$, it may personalize the structure ordering without updating the shared parameters $\mathbf{w}^*$. The adaptation data are kept separate from the final test data. Local structure importance is estimated as

$$\mathbf{a}_u^l = \Phi(\mathbf{w}^*; \mathcal{D}_u^a), \tag{3}$$

where $\Phi(\cdot)$ is the structure-importance estimation procedure used during Stage I. The local and global importance estimates are combined as

$$\mathbf{a}_u^{\text{pers}} = \lambda_u \mathbf{a}^{g,*} + (1 - \lambda_u)\mathbf{a}_u^l, \qquad \lambda_u = \frac{1}{1 + \gamma|\mathcal{D}_u^a|}, \tag{4}$$

where γ controls the influence of the adaptation-set size. Clients with limited adaptation data therefore rely more strongly on the global importance ordering. The personalized importance vector is used to construct

$$\mathcal{R}_u = \{\mathbf{m}_u^p : p \in \{0\} \cup \mathcal{P}\}, \tag{5}$$

using the same removal budgets, structured masking procedure, and layer-wise constraints as the global roadmap. Personalization therefore changes only the retained structures and does not modify $\mathbf{w}^*$. Each personalized configuration is evaluated using the local adaptation data to obtain

$$\widehat{A}_u^p = \mathrm{Acc}(\mathcal{C}(\mathbf{w}^*, \mathbf{m}_u^p); \mathcal{D}_u^a), \tag{6}$$

where $\mathcal{C}(\cdot)$ denotes physical submodel construction. The resulting estimate is stored with the personalized roadmap. Because personalized masks may retain structures from different layers, every configuration is also physically constructed and profiled before runtime use. Given the feasible set $\mathcal{F}_u(t)$ defined in Eq. (1), the client selects

$$p_u(t) = \arg\max_{p \in \mathcal{F}_u(t)} \widehat{A}_u^p. \tag{7}$$

Ties are resolved using the same lower-FLOP rule as the global runtime policy. When no adaptation data are available, the client uses

$$\mathcal{R}_u = \mathcal{R}^g, \tag{8}$$

together with the server-side validation estimates $\widehat{A}^p$. The global roadmap remains the default deployment mode because it requires no client-side adaptation, while personalization is an optional extension for clients with suitable local data.

1.3 Baseline Implementations

We adapted all baselines to the same federated training and deployment setting used by RAS-FL. The methods use identical client partitions, communication rounds, local epochs, optimizer settings, and evaluation clients.**FedAvg** and **FedProx** are implemented as full-model FL baselines. Each client trains the complete network, the server aggregates the resulting model updates, and deployment always uses the full model. FedProx additionally includes its proximal regularization term during local optimization.

Algorithm 1 Runtime-Aware Shrinkable FL (RAS-FL)

Require: Training clients $\mathcal{K}$, rounds T, local steps E, shrinkage levels $\mathcal{P}$, β, and $\rho_{\min}$
Ensure: $\mathbf{w}^*$, $\mathbf{a}^{g,*}$, and $\mathcal{R}^g$
Initialize $\mathbf{w}^0$ and $\mathbf{a}^{g,0} = \epsilon\mathbf{1}$ using Eq.(**6**)
for $t = 0, 1, \ldots, T-1$ **do**
  Select $\mathcal{S}_t \subseteq \mathcal{K}$
  if shrinkage-aware training is active **then**
    Construct $\{\mathbf{m}^{p,t} : p \in \mathcal{P}_t\}$ from $\mathbf{a}^{g,t}$ using the boundary-aware roadmap procedure
  end if
  Broadcast $\mathbf{w}^t$
  for each client $k \in \mathcal{S}_t$ **in parallel do**
    Set $\mathbf{w}_k^{t,0} \leftarrow \mathbf{w}^t$
    Compute local importance using Eq.(**7**)
    for $e = 0, 1, \ldots, E-1$ **do**
      if shrinkage-aware training is active and $z \sim \mathrm{Bernoulli}(\pi)$ yields $z = 1$ **then**
        Sample $p \in \mathcal{P}_t$
        Optimize Eq.(**16**)
        Apply Eq.(**17**)
      else
        Optimize the full-model objective (first case of Eq.(**16**))
      end if
    end for
    Upload $(\mathbf{w}_k^{t,E}, \mathbf{a}_k^t)$
  end for
  Aggregate model parameters and importance using Eq.(**10**)
  Update the global importance vector using Eq.(**11**)
  Compute S^t using Eq.(**19**)
end for
Select t^* using Eq.(**20**)
Obtain $\mathbf{w}^*$ and $\mathbf{a}^{g,*}$ using Eq.(**21**)
Construct $\mathcal{R}^g$ using Eq.(**28**)
return $\mathbf{w}^*$, $\mathbf{a}^{g,*}$, and $\mathcal{R}^g$

FedDrop-style is implemented as a random structured-submodel baseline. At each supported shrinkage level, filters, channels, or neurons are randomly deactivated while enforcing the same layer-wise minimum-capacity constraints, protected layers, and global removal budgets used by RAS-FL. This baseline therefore isolates the effect of importance-guided structure selection from the effect of structured shrinkage itself.

Following OFA-FL, **OFA-style** is implemented as a once-for-all federated baseline that jointly trains a shared model across multiple nested subnetwork configurations. At deployment, the configuration matching the requested resource budget is selected directly. Unlike RAS-FL, these subnetworks follow a predefined width or capacity schedule rather than a learned federated structure-importance roadmap.

Slimmable-NN is implemented as a single shared network with switchable-width configurations, following the federated slimmable setting of FedSNN. During local training, clients sample from the supported widths, and the server aggregates the shared

Algorithm 2 Runtime Deployment on a New Client

Require: Global model $\mathbf{w}^*$, global importance $\mathbf{a}^{g,*}$, global roadmap $\mathcal{R}^g$, profiled resource costs $\{\mathbf{c}_u^p\}_{p\in\{0\}\cup\mathcal{P}}$, estimated accuracies $\{\widehat{A}_u^p\}_{p\in\{0\}\cup\mathcal{P}}$, client u, optional adaptation data $\mathcal{D}_u^a$, and resource state $\mathbf{r}_u(t)$

Ensure: Prediction $\hat{y}$ or an infeasible-state indication

```
if D_u^a is available then
    Compute a_u^l using Eq. (3)
    Compute λ_u using Eq. (4)
    Compute a_u^pers using Eq. (4)
    Construct R_u using Eq. (5)
else
    Set R_u ← R^g
end if
Compute the feasible set F_u(t) using Eq. (1)
if F_u(t) = ∅ then
    return infeasible resource state
end if
Select p_u(t) using Eq. (29)
Retrieve m_u^{p_u(t)} from R_u
if p_u(t) differs from the currently active configuration then
    Construct w_c^{p_u(t)} ← C(w*, m_u^{p_u(t)}) on the CPU using Eq. (30)
    Release the currently active accelerator model
    Transfer w_c^{p_u(t)} to the accelerator
end if
Predict ŷ using Eq. (31)
return ŷ
```

parameters. For evaluation, each shrinkage level is matched to the closest available width configuration. Unlike RAS-FL, Slimmable-NN adapts capacity through predefined width multipliers rather than learned, importance-guided structured masks.

SLEX-style, inspired by SLEXNet and ScaleFL , is implemented as an early-exit federated baseline by attaching intermediate classifiers to a shared backbone. All exits are trained jointly during local optimization. At deployment, the exit matching the target resource budget is selected, while the final exit represents the full-depth model. Because early exits reduce computation through depth-wise termination rather than channel-level shrinkage, each exit is compared using its measured active-parameter and FLOP profile.

All methods use the same client partitions, participation schedule, communication-round budget, local epochs, base backbone, evaluation clients, and random seeds. Because the adaptive methods use different objectives and subnetwork-sampling mechanisms, their optimization hyperparameters are tuned independently using the same search budget. A held-out subset of the training-client data is used for validation and is disjoint from the final test clients and test data. Adaptive models are selected using the average validation accuracy across all supported runtime operating points, while full-model baselines are selected using full-capacity validation accuracy.

1.4 Evaluation on Sample-Disjoint Newly Introduced Clients

In addition to the official global test set, we evaluate all methods on a separate group of newly introduced deployment clients. Their samples and client-specific data distributions are not used during federated training. Before constructing any clients, the original training dataset is divided into two mutually exclusive pools:

$$\mathcal{D}_{\text{original}} = \mathcal{D}_{\text{FL}} \cup \mathcal{D}_{\text{new}}, \qquad \mathcal{D}_{\text{FL}} \cap \mathcal{D}_{\text{new}} = \emptyset. \tag{9}$$

The pool $\mathcal{D}_{\text{FL}}$ is partitioned among the federated training clients using the training Dirichlet concentration parameter. A held-out subset of this pool is reserved for validation and is used only for hyperparameter tuning and checkpoint selection; it is excluded from local model optimisation. The remaining samples are used for federated training. The held-out pool $\mathcal{D}_{\text{new}}$ is independently partitioned among 20 newly introduced clients using $\alpha = 0.08$ to induce strong label skew. No sample assigned to these clients appears in the federated training, validation, or model-selection sets. For each new client u, the local dataset is divided into two disjoint subsets:

$$\mathcal{D}_u^{\text{new}} = \mathcal{D}_u^a \cup \mathcal{D}_u^e, \qquad \mathcal{D}_u^a \cap \mathcal{D}_u^e = \emptyset, \tag{10}$$

where $\mathcal{D}_u^a$ contains 500 adaptation samples and $\mathcal{D}_u^e$ contains 500 evaluation samples. The adaptation subset is used only by methods supporting deployment-time personalisation, while the evaluation subset is never used for adaptation, tuning, or model selection. We consider two RAS-FL deployment modes:

- **Global roadmap**: the client uses the roadmap learned during federated training without local adaptation.
- **Personalised roadmap**: the client refines the global importance ordering using $\mathcal{D}_u^a$ while keeping the shared model parameters fixed.

The primary comparison with FedAvg, FedProx, OFA-style, Slimmable-NN, SLEX-style, and FedDrop-style uses the global RAS-FL roadmap, ensuring that no compared method accesses $\mathcal{D}_u^a$. Personalised RAS-FL is reported separately as an optional client-specific extension. All methods are evaluated on the same sample-disjoint deployment clients and under matched resource operating points. The code is available at **Code**.

2 Additional Experimental Results

2.1 Extended Accuracy–Efficiency Results

Table 1 provides the complete results for Fashion-MNIST, CIFAR-10, CIFAR-100, and Tiny-ImageNet-20. In addition to full-model and average runtime performance, it reports the percentages of active parameters, structures, and FLOPs relative to the corresponding full model. RAS-FL maintains the strongest runtime accuracy on CIFAR-10, CIFAR-100, and Tiny-ImageNet-20, while remaining competitive on Fashion-MNIST. The results further show that its roadmap configurations provide substantial computational savings while preserving a favourable accuracy–efficiency trade-off. Global and personalised roadmaps exhibit similar resource usage because they follow the same structural budgets.

Figure 1 further shows that the client-level accuracy distributions of RAS-FL are shifted toward higher values on the more challenging datasets. Overall, the importance-guided roadmap reduces runtime cost while preserving predictive performance.

Table 1: Extended accuracy–efficiency results on previously unseen deployment clients. Full-model and average runtime performance are reported for each method, with runtime metrics averaged over the supported shrinkage levels from 5% to 40%. Active parameters, structures, and FLOPs are expressed as percentages of the corresponding full model. Values are mean $\pm$ standard deviation over three runs.

Dataset	Method	Full Acc. (%)	Avg. Runtime Acc. (%)	Macro F1 (%)	Active Params (%)	Active Struct. (%)	Active FLOPs (%)
F-MNIST	**FedAvg**	91.79 ± 0.74	91.47 ± 0.14	91.67 ± 0.32	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	FedProx	91.83 ± 0.90	$\mathbf{91.78 \pm 0.67}$	$\mathbf{91.85 \pm 0.73}$	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	Federated Dropout-style	87.76 ± 0.22	86.56 ± 0.42	87.72 ± 0.27	72.52 ± 0.00	81.19 ± 0.00	72.00 ± 0.00
	OFA-style	91.50 ± 0.49	91.05 ± 1.13	91.12 ± 1.18	82.72 ± 0.00	83.20 ± 0.00	82.72 ± 0.00
	SLEX-style final exit	91.20 ± 1.25	91.10 ± 1.66	91.25 ± 1.72	82.73 ± 0.00	83.86 ± 0.00	82.73 ± 0.00
	Slimmable Neural Networks	90.13 ± 1.68	89.90 ± 1.82	89.95 ± 1.64	68.85 ± 0.00	82.50 ± 0.00	69.13 ± 0.00
	RAS-FL global roadmap	$\mathbf{92.19 \pm 0.44}$	91.00 ± 1.87	91.79 ± 1.14	73.02 ± 0.14	82.86 ± 0.00	73.02 ± 0.14
	RAS-FL personalized roadmap	$\mathbf{92.19 \pm 0.44}$	91.02 ± 1.91	91.57 ± 1.44	72.88 ± 0.14	82.86 ± 0.00	73.02 ± 0.14
CIFAR-10	**FedAvg**	$\mathbf{83.64 \pm 2.08}$	$\mathbf{81.53 \pm 2.15}$	$\mathbf{83.58 \pm 2.16}$	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	FedProx	81.17 ± 4.66	80.87 ± 4.76	81.03 ± 4.56	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	Federated Dropout-style	30.30 ± 13.63	34.17 ± 13.35	31.05 ± 13.98	72.64 ± 0.00	82.63 ± 0.00	71.52 ± 0.00
	OFA-style	78.19 ± 3.57	76.01 ± 3.91	76.01 ± 3.91	71.66 ± 0.00	82.83 ± 0.00	69.12 ± 0.00
	SLEX-style final exit	74.22 ± 8.29	72.35 ± 5.78	74.15 ± 8.71	75.74 ± 0.00	83.18 ± 0.00	69.34 ± 0.00
	Slimmable Neural Networks	77.11 ± 1.43	78.14 ± 0.71	77.17 ± 0.41	68.82 ± 0.00	82.50 ± 0.00	69.00 ± 0.00
	RAS-FL global roadmap	81.44 ± 3.28	80.08 ± 2.79	81.38 ± 2.23	74.62 ± 0.00	82.63 ± 0.00	78.08 ± 0.00
	RAS-FL personalized roadmap	81.44 ± 3.28	79.85 ± 3.32	81.45 ± 3.38	74.62 ± 0.00	82.63 ± 0.00	78.08 ± 0.00
CIFAR-100	**FedAvg**	57.59 ± 5.71	56.37 ± 5.89	57.18 ± 5.76	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	FedProx	49.73 ± 4.66	49.11 ± 4.62	49.79 ± 4.61	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	Federated Dropout-style	11.84 ± 2.90	8.32 ± 0.72	8.32 ± 0.72	68.84 ± 0.36	82.50 ± 0.00	70.20 ± 0.55
	OFA-style	18.63 ± 7.67	14.15 ± 3.64	14.15 ± 3.64	67.26 ± 1.33	82.50 ± 0.00	67.28 ± 1.19
	SLEX-style final exit	24.02 ± 8.91	22.17 ± 14.62	22.17 ± 14.62	68.01 ± 2.07	82.50 ± 0.00	68.13 ± 1.05
	Slimmable Neural Networks	39.12 ± 6.25	40.75 ± 7.04	40.75 ± 7.04	68.79 ± 0.00	82.50 ± 0.00	68.88 ± 0.00
	RAS-FL global roadmap	$\mathbf{64.20 \pm 2.43}$	$\mathbf{54.95 \pm 1.08}$	$\mathbf{57.91 \pm 2.13}$	78.52 ± 0.28	82.50 ± 0.00	68.81 ± 0.47
	RAS-FL personalized roadmap	$\mathbf{64.20 \pm 2.43}$	54.22 ± 1.89	56.61 ± 1.40	78.52 ± 0.28	82.50 ± 0.00	68.81 ± 0.47
Tiny-IN20	**FedAvg**	33.10 ± 13.87	32.18 ± 14.53	33.81 ± 14.07	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	FedProx	$\mathbf{43.47 \pm 1.31}$	$\mathbf{43.71 \pm 1.55}$	$\mathbf{43.45 \pm 1.26}$	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	Federated Dropout-style	5.64 ± 3.00	4.83 ± 2.85	4.83 ± 2.85	85.54 ± 0.00	82.50 ± 0.08	86.05 ± 0.03
	OFA-style	9.91 ± 3.39	12.27 ± 3.93	12.27 ± 3.93	86.83 ± 0.00	82.50 ± 0.00	87.30 ± 0.00
	SLEX-style final exit	13.46 ± 2.35	17.12 ± 6.40	17.12 ± 6.40	86.86 ± 0.00	82.50 ± 0.00	87.24 ± 0.13
	Slimmable Neural Networks	6.60 ± 1.98	7.49 ± 1.94	7.49 ± 1.94	86.24 ± 0.00	82.50 ± 0.00	86.53 ± 0.20
	RAS-FL global roadmap	38.33 ± 1.65	33.14 ± 3.13	33.14 ± 3.13	89.07 ± 0.00	82.50 ± 0.00	89.64 ± 0.07
	RAS-FL personalized roadmap	37.98 ± 2.04	32.97 ± 3.80	32.97 ± 3.80	89.07 ± 0.00	82.50 ± 0.00	89.64 ± 0.07

Accuracy–FLOP Trade-off Figure 2 complements the active parameter analysis by comparing runtime accuracy with average active FLOPs. RAS-FL provides a favourable accuracy–computation trade-off across all datasets. The advantage is particularly clear on CIFAR-100 and Tiny-ImageNet-200, where RAS-FL substantially outperforms the adaptive baselines at comparable or lower computational cost. On Fashion MNIST and CIFAR-10, it preserves high accuracy while reducing the active computation relative to the full model.

2.2 Training Convergence

Figure 3 shows that RAS-FL converges smoothly on Fashion-MNIST and CIFAR-10 and maintains a stable upward trend on CIFAR-100. Tiny-ImageNet-200 exhibits larger round-to-round fluctuations due to its greater task complexity and non-IID client distributions; however, RAS-FL continues improving without divergence. Overall, shrinkage-aware training preserves reliable convergence while jointly preparing the full model and runtime submodels.

2.3 Shrinkage-Aware Training Dynamics

Figure 4 compares full-model training accuracy with masked-submodel validation accuracy as the supported mask schedule expands. Introducing additional shrinkage levels can temporarily reduce masked accuracy, particularly on the more challenging datasets, but performance subsequently recovers as the model adapts to the expanded runtime capacities. This indicates that shrinkage-aware training improves submodel robustness without destabilizing the full model.

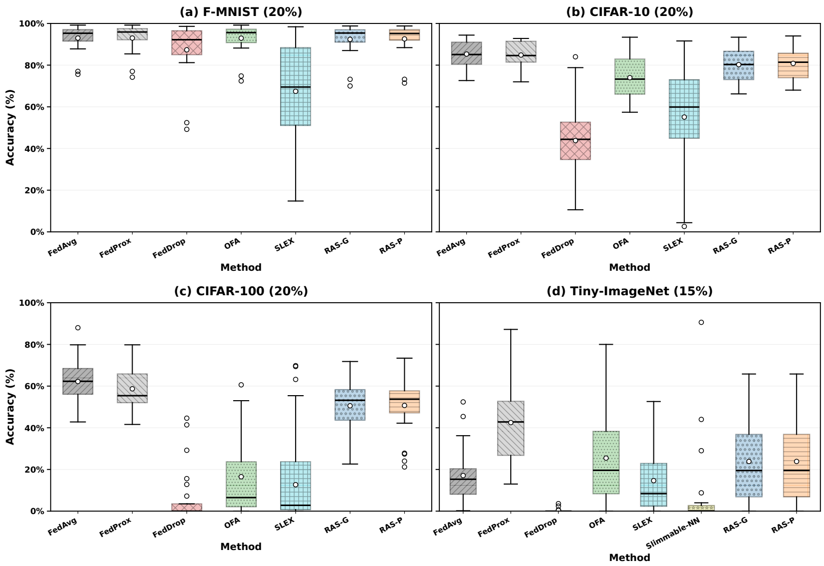

Figure 1: Client-level runtime accuracy at representative shrinkage levels: 30% for Fashion-MNIST and the CIFAR datasets, and 15% for Tiny-ImageNet-200.

2.4 Per-Level Runtime Adaptation Results

Table 2 shows that the useful shrinkage range is dataset- and architecture-dependent. Accuracy generally remains stable over moderate removal levels, after which a dataset-specific degradation point appears. Fashion-MNIST remains robust even under aggressive shrinkage, while CIFAR-10 preserves most of its accuracy through approximately 30% removal. The more challenging CIFAR-100 and Tiny-ImageNet-20 settings exhibit earlier and sharper degradation, reflecting their greater task and representation complexity. For CIFAR-100, the global roadmap retains $51.60 \pm 2.38\%$ accuracy at 30% structure removal, while physical compaction reduces parameters by $58.61 \pm 0.17\%$, FLOPs by $53.49 \pm 2.01\%$, compact-state size by $58.60 \pm 0.17\%$, and model memory by $56.49 \pm 0.34\%$.The larger drop at 40% suggests that this shrinkage level removes more capacity than the trained model can tolerate on CIFAR-100. This degradation is not unexpected under aggressive structured shrinkage, particularly because no additional fine-tuning or retraining is performed after the compact submodel is selected and physically constructed. The roadmap should therefore be viewed as a set of operating points that provide different accuracy–resource trade-offs, rather than as configurations that are all expected to preserve the same accuracy. Importantly, switching between these operating points requires neither retraining nor deployment-time fine tuning. Each compact model is reconstructed from the fixed global model using its precomputed roadmap, making runtime adaptation lightweight and reversible.

Sensitivity to checkpoint-selection weights. Table 5 shows that Fashion-MNIST is relatively insensitive to the checkpoint weights, although ignoring masked-submodel performance, i.e., $(1.00, 0.00)$, yields the lowest average runtime accuracy. CIFAR-10 is more sensitive, with the balanced setting $(0.50, 0.50)$ providing the highest and most stable accuracy across all shrinkage levels. In contrast, assigning greater weight to

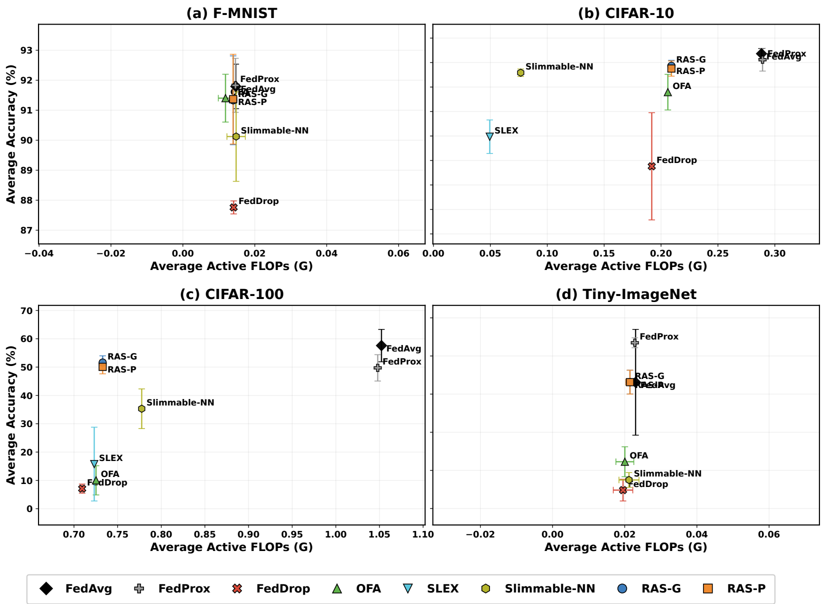

Figure 2: Runtime accuracy versus average active FLOPs on previously unseen clients. Error bars indicate standard deviation over three independent runs.

the masked-submodel term, $(0.25, 0.75)$, substantially reduces accuracy and increases variance. This suggests that, for the more challenging CIFAR-10 task, the full-model term should not be underweighted. Because the full model provides the shared representation from which all compact subnetworks are derived, overemphasising masked performance may favour checkpoints specialised to sampled subnetworks at the expense of general representation quality. Overall, the results support balanced consideration of full-model and masked-submodel accuracy during checkpoint selection.

Sensitivity to the number of training shrinkage levels. Table 6 evaluates the number of shrinkage levels sampled during shrinkage-aware training. Overall, RAS-FL shows limited sensitivity to this hyperparameter. On Fashion-MNIST, the full-model accuracy varies by only 0.21 percentage points across all configurations. On CIFAR-10, the variation is slightly larger, with the best and worst configurations differing by about one percentage point. Although the two- and five-level settings perform marginally better on Fashion-MNIST and the three-level setting achieves the highest accuracy on CIFAR-10, the seven-level configuration used by RAS-FL remains competitive while providing finer coverage of runtime capacities. These results indicate that increasing the number of training shrinkage levels incurs only a small reduction in full-model accuracy.

Sensitivity to the mask-training probability. Table 7 shows that the effect of p_{mask} is task dependent. On Fashion-MNIST, lower masking probabilities perform best: $p_{\text{mask}} = 0.10$ achieves the highest accuracy, macro-F1, balanced accuracy, and checkpoint score, while $p_{\text{mask}} = 0.25$ provides nearly identical performance. Increasing p_{mask} to 0.50 or 0.75 progressively reduces accuracy and leads to earlier checkpoint selection, suggesting that excessive masked updates can weaken full-model optimisa-

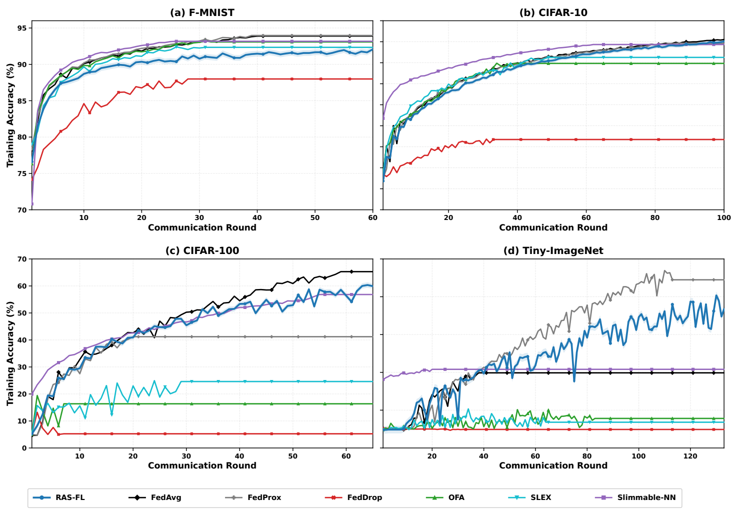

Figure 3: Training convergence of RAS-FL and competing baselines over communication rounds. Despite jointly training the full model and runtime-adaptive submodels, RAS-FL exhibits stable convergence across datasets.

tion.

CIFAR-10 benefits from a moderate masking frequency. $p_{\text{mask}} = 0.25$ provides the highest test accuracy and lowest loss, whereas $p_{\text{mask}} = 0.50$ achieves the highest checkpoint score with low variance. By contrast, $p_{\text{mask}} = 0.10$ exhibits greater variability, suggesting insufficient exposure to compact submodels, while $p_{\text{mask}} = 0.75$ reduces predictive performance. Overall, values between 0.25 and 0.50 provide the best balance between full-model learning and robustness across compact subnetworks.

Sensitivity to the importance-score EMA coefficient. Table 8 shows that Fashion-MNIST is relatively insensitive to β, with all settings achieving similar predictive performance. The setting $\beta = 0.95$ provides the highest accuracy, macro-F1, and balanced accuracy, whereas $\beta = 0.90$ gives the highest checkpoint score and $\beta = 0.99$ the lowest test loss. On CIFAR-10, larger EMA coefficients generally provide more stable importance estimates: $\beta = 0.99$ achieves the highest accuracy with low variance, while $\beta = 0.95$ obtains the highest checkpoint score. In contrast, $\beta = 0.90$ substantially reduces accuracy and increases variability. Overall, $\beta \in \{0.95, 0.99\}$ provides the most reliable balance between smoothing short-term importance fluctuations and remaining responsive to changes during federated training.

Future Work. The present evaluation focuses on the physical resource footprints of the supported configurations, including parameters, FLOPs, compact-state size, model memory, and peak memory. End-to-end configuration-switching latency is not evaluated and is left for future deployment studies.

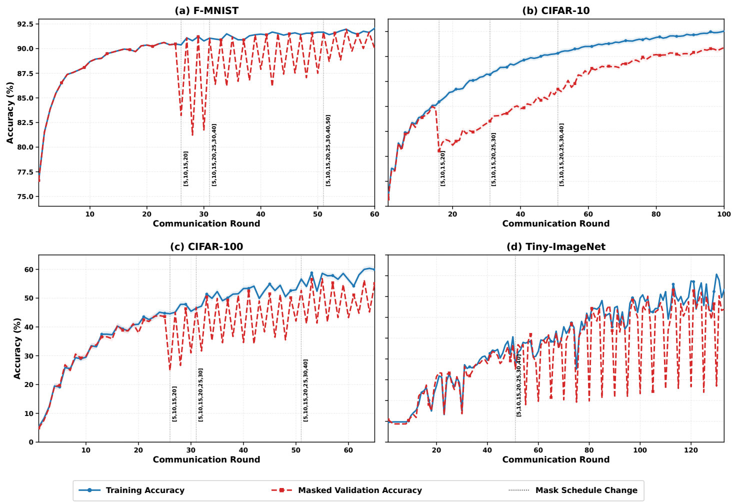

Figure 4: Shrinkage-aware training dynamics across communication rounds. Vertical lines mark expansions of the mask schedule, while the curves compare full-model training accuracy with masked-submodel validation accuracy.

Table 2: Runtime accuracy (%) at different shrinkage levels on previously unseen clients. Values are mean $\pm$ standard deviation over three runs.

Dataset	Method	0%	10%	20%	30%	40%	50%
F-MNIST	**FedAvg**	91.79 ± 0.74	N/A	N/A	N/A	N/A	N/A
	FedProx	91.83 ± 0.90	N/A	N/A	N/A	N/A	N/A
	Federated Dropout-style	87.76 ± 0.22	87.02 ± 3.47	86.81 ± 1.15	84.33 ± 1.60	83.56 ± 2.82	83.92 ± 1.24
	OFA-style	91.50 ± 0.49	$\mathbf{91.58 \pm 0.62}$	90.82 ± 1.48	90.43 ± 1.51	89.82 ± 1.03	88.51 ± 1.40
	SLEX-style final exit	91.20 ± 1.25	91.27 ± 1.62	$\mathbf{91.05 \pm 1.79}$	$\mathbf{90.88 \pm 1.74}$	89.49 ± 2.80	87.02 ± 4.22
	Slimmable-NN	90.13 ± 1.68	90.17 ± 1.43	90.05 ± 1.80	89.22 ± 2.80	89.37 ± 1.75	88.74 ± 2.12
	RAS-FL global roadmap	$\mathbf{92.19 \pm 0.44}$	91.00 ± 2.06	90.93 ± 1.98	90.71 ± 2.03	90.26 ± 1.43	$\mathbf{89.97 \pm 0.25}$
	RAS-FL personalized roadmap	$\mathbf{92.19 \pm 0.44}$	91.03 ± 2.12	91.03 ± 2.00	90.61 ± 2.14	$\mathbf{90.36 \pm 1.17}$	89.26 ± 0.33
CIFAR-10	**FedAvg**	$\mathbf{83.64 \pm 2.08}$	N/A	N/A	N/A	N/A	N/A
	FedProx	81.17 ± 4.66	N/A	N/A	N/A	N/A	N/A
	Federated Dropout-style	30.30 ± 13.63	34.07 ± 7.80	36.97 ± 16.37	37.63 ± 21.90	31.12 ± 16.54	17.12 ± 5.65
	OFA-style	78.19 ± 3.57	80.14 ± 2.73	77.17 ± 4.01	67.92 ± 7.28	54.59 ± 4.58	27.59 ± 5.36
	SLEX-style final exit	74.22 ± 8.29	75.60 ± 6.77	73.37 ± 5.16	65.94 ± 4.89	47.22 ± 5.66	24.79 ± 9.99
	Slimmable-NN	77.11 ± 1.43	78.40 ± 1.23	78.97 ± 1.07	75.85 ± 1.51	65.08 ± 1.32	57.93 ± 4.01
	RAS-FL global roadmap	81.44 ± 3.28	80.74 ± 3.44	$\mathbf{80.38 \pm 2.87}$	$\mathbf{78.68 \pm 2.25}$	$\mathbf{72.12 \pm 2.10}$	$\mathbf{69.02 \pm 1.96}$
	RAS-FL personalized roadmap	81.44 ± 3.28	$\mathbf{80.86 \pm 3.59}$	79.92 ± 3.53	77.55 ± 3.11	71.62 ± 3.03	67.66 ± 3.36
CIFAR-100	**FedAvg**	57.59 ± 5.71	N/A	N/A	N/A	N/A	N/A
	FedProx	49.73 ± 4.66	N/A	N/A	N/A	N/A	N/A
	Federated Dropout-style	11.84 ± 2.90	9.64 ± 0.60	8.85 ± 0.37	7.09 ± 1.61	6.68 ± 1.74	5.70 ± 3.00
	OFA-style	18.63 ± 7.67	16.75 ± 5.20	13.03 ± 2.67	10.03 ± 5.15	5.14 ± 7.86	2.41 ± 2.09
	SLEX-style final exit	24.02 ± 8.91	26.70 ± 14.28	23.88 ± 20.90	12.27 ± 9.01	8.38 ± 5.33	7.72 ± 3.88
	Slimmable-NN	39.12 ± 6.25	39.89 ± 8.16	43.76 ± 8.33	35.30 ± 7.01	$\mathbf{21.11 \pm 10.12}$	$\mathbf{10.77 \pm 6.65}$
	RAS-FL global roadmap	$\mathbf{64.20 \pm 2.43}$	$\mathbf{60.88 \pm 2.13}$	$\mathbf{60.14 \pm 1.48}$	$\mathbf{51.60 \pm 2.38}$	17.78 ± 8.43	7.81 ± 1.53
	RAS-FL personalized roadmap	$\mathbf{64.20 \pm 2.43}$	60.70 ± 2.16	59.66 ± 2.31	50.07 ± 2.44	17.61 ± 5.68	5.15 ± 1.70
Tiny-IN20	**FedAvg**	33.10 ± 13.87	N/A	N/A	N/A	N/A	N/A
	FedProx	$\mathbf{43.47 \pm 1.31}$	N/A	N/A	N/A	N/A	N/A
	Federated Dropout-style	5.64 ± 3.00	4.43 ± 3.41	6.22 ± 3.51	3.01 ± 2.36	N/A	N/A
	OFA-style	16.54 ± 7.85	12.34 ± 2.57	10.28 ± 2.73	9.91 ± 3.39	N/A	N/A
	SLEX-style final exit	20.08 ± 12.02	18.19 ± 7.02	13.46 ± 2.35	12.74 ± 5.48	N/A	N/A
	Slimmable-NN	9.09 ± 0.62	7.28 ± 3.14	6.97 ± 2.56	6.60 ± 1.98	N/A	N/A
	RAS-FL global roadmap	38.33 ± 1.65	$\mathbf{37.00 \pm 3.50}$	$\mathbf{34.52 \pm 3.09}$	$\mathbf{22.70 \pm 7.78}$	N/A	N/A
	RAS-FL personalized roadmap	37.98 ± 2.04	36.00 ± 4.02	32.52 ± 5.56	21.60 ± 6.43	N/A	N/A

Table 3: Physical compression and server-GPU runtime of compact RAS-FL across 20 clients. Values are mean ± standard deviation over three independent runs.

Data	Variant	Metric	Full	10%	20%	30%	40%
F-MNIST	Global	**Accuracy (%)**	91.62 ± 0.01	90.88 ± 0.67	90.91 ± 0.73	90.80 ± 0.76	89.97 ± 0.25
		Parameter reduction (%)	0.00 ± 0.00	14.71 ± 0.00	32.44 ± 0.00	50.21 ± 0.05	67.63 ± 0.00
		FLOPs reduction (%)	0.00 ± 0.00	14.62 ± 0.00	17.95 ± 0.00	24.90 ± 0.22	44.45 ± 0.00
		Compact-state reduction (%)	0.00 ± 0.00	14.71 ± 0.00	32.43 ± 0.00	50.20 ± 0.05	67.62 ± 0.00
		Model-memory reduction (%)	0.00 ± 0.00	6.10 ± 0.00	32.39 ± 0.00	50.14 ± 0.04	67.54 ± 0.00
		Peak-memory reduction (%)	0.00 ± 0.00	1.30 ± 0.00	6.91 ± 0.00	10.70 ± 0.01	14.44 ± 0.00
	Personalised	**Accuracy (%)**	91.62 ± 0.01	90.92 ± 0.73	91.14 ± 0.61	91.24 ± 0.73	89.91 ± 0.52
		Parameter reduction (%)	0.00 ± 0.00	14.71 ± 0.00	32.43 ± 0.01	50.07 ± 0.03	67.62 ± 0.21
		FLOPs reduction (%)	0.00 ± 0.00	14.62 ± 0.00	17.97 ± 0.04	24.69 ± 0.43	44.46 ± 0.29
		Compact-state reduction (%)	0.00 ± 0.00	14.71 ± 0.00	32.42 ± 0.01	50.06 ± 0.03	67.61 ± 0.21
		Model-memory reduction (%)	0.00 ± 0.00	6.10 ± 0.00	32.38 ± 0.01	50.00 ± 0.04	67.53 ± 0.21
		Peak-memory reduction (%)	0.00 ± 0.00	1.30 ± 0.00	6.91 ± 0.00	10.67 ± 0.01	14.44 ± 0.04
CIFAR-10	Global	**Accuracy (%)**	82.57 ± 1.85	82.70 ± 2.02	81.56 ± 0.86	81.21 ± 0.23	75.06 ± 2.81
		Parameter reduction (%)	0.00 ± 0.00	19.16 ± 1.21	40.23 ± 1.43	57.46 ± 2.17	68.71 ± 0.08
		FLOPs reduction (%)	0.00 ± 0.00	15.22 ± 0.81	26.54 ± 2.02	34.11 ± 1.41	42.81 ± 0.17
		Compact-state reduction (%)	0.00 ± 0.00	19.15 ± 1.21	40.22 ± 1.43	57.45 ± 2.17	68.69 ± 0.08
		Model-memory reduction (%)	0.00 ± 0.00	20.06 ± 0.11	41.65 ± 1.37	54.69 ± 1.45	69.53 ± 1.19
		Peak-memory reduction (%)	0.00 ± 0.00	5.94 ± 0.02	11.29 ± 0.44	15.17 ± 0.36	18.73 ± 0.82
	Personalised	**Accuracy (%)**	82.57 ± 1.85	82.46 ± 1.68	82.02 ± 1.25	80.15 ± 0.26	74.79 ± 0.50
		Parameter reduction (%)	0.00 ± 0.00	18.86 ± 1.02	39.62 ± 1.28	57.34 ± 1.44	68.69 ± 0.07
		FLOPs reduction (%)	0.00 ± 0.00	14.40 ± 1.25	25.98 ± 1.27	34.90 ± 0.93	43.23 ± 0.01
		Compact-state reduction (%)	0.00 ± 0.00	18.86 ± 1.02	39.61 ± 1.28	57.33 ± 1.44	68.67 ± 0.07
		Model-memory reduction (%)	0.00 ± 0.00	19.02 ± 0.75	41.27 ± 1.23	54.80 ± 1.79	69.51 ± 1.20
		Peak-memory reduction (%)	0.00 ± 0.00	5.52 ± 0.07	10.53 ± 0.49	15.35 ± 0.65	18.73 ± 0.82
CIFAR-100	Global	**Accuracy (%)**	64.20 ± 2.43	60.88 ± 2.13	60.14 ± 1.48	51.60 ± 2.38	17.78 ± 8.43
		Parameter reduction (%)	0.00 ± 0.00	19.00 ± 0.44	40.32 ± 0.90	58.61 ± 0.17	66.88 ± 0.03
		FLOPs reduction (%)	0.00 ± 0.00	22.42 ± 0.59	38.67 ± 0.07	53.49 ± 2.01	63.70 ± 0.31
		Compact-state reduction (%)	0.00 ± 0.00	19.00 ± 0.44	40.31 ± 0.90	58.60 ± 0.17	66.87 ± 0.03
		Model-memory reduction (%)	0.00 ± 0.00	22.79 ± 0.42	43.07 ± 0.86	56.49 ± 0.34	68.39 ± 0.03
		Peak-memory reduction (%)	0.00 ± 0.00	3.97 ± 0.00	13.85 ± 0.88	20.76 ± 0.99	24.44 ± 0.73
	Personalised	**Accuracy (%)**	64.20 ± 2.43	60.70 ± 2.16	59.66 ± 2.31	50.07 ± 2.44	17.61 ± 5.68
		Parameter reduction (%)	0.00 ± 0.00	18.86 ± 0.43	40.18 ± 0.60	58.69 ± 0.11	66.88 ± 0.02
		FLOPs reduction (%)	0.00 ± 0.00	22.32 ± 0.62	38.54 ± 0.23	52.58 ± 1.28	63.70 ± 0.20
		Compact-state reduction (%)	0.00 ± 0.00	18.86 ± 0.43	40.17 ± 0.60	58.68 ± 0.11	66.87 ± 0.02
		Model-memory reduction (%)	0.00 ± 0.00	22.66 ± 0.41	42.94 ± 0.57	56.35 ± 0.23	68.39 ± 0.02
		Peak-memory reduction (%)	0.00 ± 0.00	4.60 ± 0.39	14.17 ± 1.02	20.58 ± 0.86	24.22 ± 1.01

Table 4: Complete ablation accuracy (%) on previously unseen Fashion-MNIST and CIFAR-10 clients. Values are mean ± standard deviation over three runs. The highest value in each column is shown in bold with grey shading.

Dataset	Variant	Full	5%	10%	15%	20%	25%	30%	40%
F-MNIST	**Full RAS-FL**	92.03 ± 0.23	91.43 ± 1.32	90.92 ± 2.21	90.89 ± 2.08	90.90 ± 2.03	**91.18 ± 1.99**	**91.04 ± 1.74**	**90.52 ± 0.69**
	Random masks	91.91 ± 1.01	91.47 ± 1.35	90.93 ± 1.67	90.03 ± 1.22	87.61 ± 4.52	88.18 ± 2.60	87.38 ± 1.75	79.96 ± 3.28
	No distillation	92.01 ± 0.91	91.25 ± 1.19	**91.23 ± 1.17**	**91.21 ± 1.22**	**91.19 ± 1.19**	90.10 ± 1.31	89.84 ± 1.42	87.52 ± 2.02
	No shrink training	92.04 ± 0.72	91.02 ± 0.80	89.52 ± 2.40	89.11 ± 2.50	88.16 ± 2.98	86.67 ± 3.05	84.52 ± 3.40	76.52 ± 5.83
	No runtime selection	**92.31 ± 0.80**	**91.78 ± 1.48**	91.15 ± 2.32	91.08 ± 2.30	90.98 ± 2.36	91.09 ± 2.46	90.91 ± 2.13	89.87 ± 2.20
CIFAR-10	**Full RAS-FL**	82.67 ± 2.02	**82.60 ± 2.03**	**82.36 ± 2.30**	**81.83 ± 2.07**	**81.16 ± 2.15**	**81.02 ± 2.07**	**80.21 ± 3.38**	**72.71 ± 4.01**
	Random masks	72.63 ± 11.75	70.09 ± 14.09	63.77 ± 23.13	59.54 ± 22.11	50.01 ± 20.37	41.23 ± 15.09	38.16 ± 11.49	20.13 ± 6.46
	No distillation	**82.95 ± 2.10**	82.57 ± 2.33	82.33 ± 2.16	81.71 ± 2.03	80.34 ± 2.20	80.80 ± 2.23	74.72 ± 3.10	66.98 ± 2.51
	No shrink training	82.31 ± 2.89	81.05 ± 2.30	79.88 ± 3.16	77.79 ± 4.16	75.48 ± 4.18	72.99 ± 4.70	68.65 ± 4.36	58.27 ± 4.67
	No runtime selection	80.09 ± 4.69	79.87 ± 5.15	79.52 ± 4.91	79.00 ± 4.56	78.01 ± 4.98	76.56 ± 4.76	75.11 ± 4.39	63.92 ± 5.22

Table 5: Sensitivity analysis of the runtime-aware checkpoint weights on newly introduced Fashion-MNIST and CIFAR-10 clients. Values are mean ± standard deviation over three independent runs.

Data	$(\lambda_{\text{full}}, \lambda_{\text{mask}})$	Full Acc.	Runtime Avg.	10%	20%	30%	40%	Active Params	Active Struct.	Active FLOPs	Runs
F-MNIST	$(\mathbf{1.00}, \mathbf{0.00})$	92.90 ± 0.37	92.31 ± 0.29	92.53 ± 0.20	92.54 ± 0.20	92.29 ± 0.38	91.90 ± 0.60	64.39 ± 0.14	75.51 ± 0.00	64.39 ± 0.14	3
	$(\mathbf{0.75}, \mathbf{0.25})$	**92.93 ± 0.38**	**92.38 ± 0.27**	**92.61 ± 0.27**	92.49 ± 0.29	92.34 ± 0.35	**92.08 ± 0.65**	64.39 ± 0.14	75.51 ± 0.00	64.39 ± 0.14	3
	$(\mathbf{0.50}, \mathbf{0.50})$	**92.93 ± 0.38**	92.37 ± 0.26	**92.61 ± 0.25**	92.54 ± 0.33	**92.40 ± 0.35**	91.95 ± 0.46	64.39 ± 0.13	75.51 ± 0.00	64.39 ± 0.13	3
	$(\mathbf{0.25}, \mathbf{0.75})$	**92.93 ± 0.38**	92.36 ± 0.30	**92.61 ± 0.27**	**92.56 ± 0.31**	92.34 ± 0.39	91.93 ± 0.72	64.40 ± 0.14	75.51 ± 0.00	64.40 ± 0.14	3
CIFAR-10	$(\mathbf{1.00}, \mathbf{0.00})$	81.05 ± 1.09	77.31 ± 1.26	80.55 ± 1.22	79.89 ± 0.82	77.59 ± 1.05	71.22 ± 2.43	70.91 ± 0.93	75.18 ± 0.00	70.91 ± 0.93	3
	$(\mathbf{0.75}, \mathbf{0.25})$	77.77 ± 4.30	74.73 ± 3.93	77.06 ± 3.97	77.52 ± 3.34	74.61 ± 4.17	69.74 ± 4.65	73.17 ± 2.66	75.18 ± 0.00	73.17 ± 2.66	3
	$(\mathbf{0.50}, \mathbf{0.50})$	**82.16 ± 0.26**	**79.29 ± 1.27**	**81.54 ± 0.67**	**81.31 ± 0.71**	**79.85 ± 1.07**	**74.44 ± 2.82**	69.93 ± 0.22	75.18 ± 0.00	69.93 ± 0.22	3
	$(\mathbf{0.25}, \mathbf{0.75})$	74.38 ± 12.51	67.25 ± 21.32	74.98 ± 11.25	71.33 ± 16.91	65.18 ± 25.68	57.53 ± 31.42	73.40 ± 5.73	75.18 ± 0.00	73.40 ± 5.73	3

Table 6: Sensitivity to the number of shrinkage levels sampled during shrinkage-aware training on Fashion-MNIST and CIFAR-10. Values are mean $\pm$ standard deviation over three independent runs.

Data	No.	Training shrinkage levels (%)	Full Acc. (%)	Selected Round
F-MNIST	2	$\{10, 40\}$	$\mathbf{91.96 \pm 0.19}$	49.67 ± 3.06
	3	$\{10, 25, 40\}$	91.75 ± 0.16	43.67 ± 5.03
	5	$\{5, 10, 20, 30, 40\}$	91.95 ± 0.04	49.67 ± 7.02
	7	$\{5, 10, 15, 20, 25, 30, 40\}$	91.80 ± 0.05	45.00 ± 2.00
CIFAR-10	2	$\{10, 40\}$	79.35 ± 1.25	92.00 ± 10.39
	3	$\{10, 25, 40\}$	$\mathbf{80.45 \pm 0.28}$	99.33 ± 1.15
	5	$\{5, 10, 20, 30, 40\}$	80.30 ± 0.18	99.33 ± 1.15
	7	$\{5, 10, 15, 20, 25, 30, 40\}$	79.45 ± 1.41	90.00 ± 13.00

Table 7: Sensitivity analysis of the mask-training probability p_{mask} on Fashion-MNIST and CIFAR-10 using their corresponding CNN architectures. Values are mean $\pm$ standard deviation over three independent runs. Accuracy, macro-F1, balanced accuracy, and checkpoint score as percentages.

Data	p_{mask}	Test Acc.	Macro-F1	Balanced Acc.	Test Loss	Checkpoint Score	Best Round	Runs
F-MNIST	0.10	$\mathbf{91.95 \pm 0.08}$	$\mathbf{91.93 \pm 0.05}$	$\mathbf{91.82 \pm 0.12}$	$\mathbf{0.2294 \pm 0.0017}$	$\mathbf{92.58 \pm 0.27}$	49.67 ± 3.06	3
	0.25	91.93 ± 0.17	91.92 ± 0.54	91.63 ± 0.33	0.2299 ± 0.0030	92.52 ± 0.30	48.33 ± 3.06	3
	0.50	91.38 ± 0.59	91.36 ± 0.28	90.78 ± 0.09	0.2386 ± 0.0135	92.21 ± 0.31	39.67 ± 14.47	3
	0.75	90.99 ± 0.33	90.96 ± 0.31	90.23 ± 0.29	0.2500 ± 0.0037	91.92 ± 0.13	24.00 ± 1.00	3
CIFAR-10	0.10	79.46 ± 3.05	79.44 ± 3.03	78.60 ± 2.82	0.5981 ± 0.0847	73.85 ± 5.03	86.00 ± 22.54	3
	0.25	$\mathbf{79.55 \pm 0.73}$	$\mathbf{79.54 \pm 0.71}$	$\mathbf{79.02 \pm 0.69}$	$\mathbf{0.5929 \pm 0.0201}$	76.43 ± 0.73	91.67 ± 8.50	3
	0.50	79.21 ± 0.35	79.18 ± 0.31	78.29 ± 0.21	0.6058 ± 0.0040	$\mathbf{77.31 \pm 0.43}$	98.67 ± 1.15	3
	0.75	77.49 ± 0.57	77.45 ± 0.55	77.40 ± 0.82	0.6563 ± 0.0186	76.65 ± 0.24	99.00 ± 1.00	3

Table 8: Sensitivity analysis of the importance-score exponential moving-average coefficient β on Fashion-MNIST and CIFAR-10. Values are mean $\pm$ standard deviation over three independent runs. Accuracy, macro-F1, balanced accuracy, and checkpoint score as percentages.

Data	EMA β	Test Acc.	Macro-F1	Balanced Acc.	Test Loss	Checkpoint Score	Best Round	Runs
F-MNIST	0.80	91.83 ± 0.03	91.15 ± 0.21	90.73 ± 0.27	0.2298 ± 0.0026	92.59 ± 0.12	49.67 ± 8.33	3
	0.90	91.83 ± 0.05	91.15 ± 0.22	90.77 ± 0.19	0.2297 ± 0.0022	$\mathbf{92.61 \pm 0.12}$	49.00 ± 7.21	3
	0.95	$\mathbf{91.93 \pm 0.17}$	$\mathbf{91.89 \pm 0.25}$	$\mathbf{91.23 \pm 0.11}$	0.2299 ± 0.0030	92.52 ± 0.30	48.33 ± 3.06	3
	0.99	91.89 ± 0.13	91.91 ± 0.06	90.91 ± 0.41	$\mathbf{0.2292 \pm 0.0042}$	92.52 ± 0.15	46.33 ± 3.06	3
CIFAR-10	0.80	79.47 ± 1.20	79.49 ± 1.26	76.21 ± 1.29	$\mathbf{0.5770 \pm 0.0166}^{\dagger}$	76.24 ± 1.24	91.33 ± 9.02	3
	0.90	75.57 ± 5.93	75.44 ± 5.66	74.87 ± 4.36	$0.7372 \pm 0.2080^{\dagger}$	71.38 ± 7.93	69.67 ± 29.50	3
	0.95	79.55 ± 0.73	79.57 ± 0.62	77.78 ± 0.98	$0.5820 \pm 0.0097^{\dagger}$	$\mathbf{76.43 \pm 0.73}$	91.67 ± 8.50	3
	0.99	$\mathbf{79.93 \pm 0.45}$	$\mathbf{79.96 \pm 0.42}$	$\mathbf{79.03 \pm 0.25}$	$0.5906 \pm 0.0187^{\dagger}$	76.06 ± 0.50	96.00 ± 5.29	3